Academic Performance Analytics for AIH

Capstone Project: CUSTOMER & MARKET INSIGHTS (DATA ANALYTICS)

Client Details

Name	Email	Phone	Preferred Contact Method
MOHAMMED AL-ZOBBI	m.al-Zobbi@aih.edu.au	0405048886	Via Email and Zoom

Team Details

Name	Email	Phone	Role
DIWANSHU HIRANWAL	Diwanshupal1@gmail.com	0449922224	Dashboard & Visualization Designer

DIYA GROVER	Diyagrover0017@gmail.com	0426249179	Team Leader & Data Visualization Lead
SOFIYAN	232114@students.aih.edu.au	0426591765	Research & Data Analyst
JOYAB ALAM	230586@students.aih.edu.au	0414918047	Documentation & Quality Coordinator

Practera

Project Background

- What challenge is the client experiencing?

AIH experiences significant student attrition due to academic underperformance, specifically high rates of FNE (Fail Not Engaged) and FNS (Fail Not Submitted) cases. This is compounded by intense competition from other colleges and government immigration caps limiting international student intake.

- Why is this issue important or urgent to address?

Addressing this issue is urgent for maintaining enrollment numbers, protecting institutional reputation, and ensuring financial sustainability. Improving student outcomes directly impacts retention rates and reduces revenue loss from student dropouts.

- How does it connect to the assigned capstone project?

This Customer & Market Insights project leverages data analytics to transform raw student performance data into actionable intelligence. By analyzing mark patterns and engagement metrics, we can uncover root causes of poor performance and attrition.

- What business or user needs are being solved?

. Identify early warning signs of at-risk students for proactive intervention

· Understand performance trends across different courses and assessment types

· Develop data-driven strategies to improve teaching methods and student support
· Enhance marketing targeting by understanding which student segments succeed
· Create evidence-based interventions to reduce failure rates and improve retention
· Optimize resource allocation to areas with greatest impact on student success

Project Scope

Deliverables:

1. A thorough report on student performance that includes mark distribution analysis, FNE/FNS trends, assessment correlation, and identification of student cohorts at risk.
2. An interactive Power BI dashboard prototype with trend identification tools, comparative analysis (with and excluding FNE/FNS), and filterable student performance metrics displays.
3. Strategic Recommendations for Academic Improvement, which include proposals for curriculum modifications, early warning system design, and data-supported intervention techniques.

Within Scope:

A thorough examination of A1, A2, A3, and overall grades for every course Thorough classification and pattern analysis of FNE/FNS creation of an interactive Power BI dashboard using KPIs Comparing with external education datasets that are readily available Finding at-risk student profiles and performance patterns A set of recommendations for academic interventions

Project Outcomes	Key activities to achieve the outcome	Activity lead
Clear understanding of client requirements	Student performance analysis and pattern identification.	DIYA GROVER
Performance of benchmarking and comparative analysis	Analyze performance variation and identify improvement areas.	SOFIYA
Interactive Powerbi dashboard development	Ensuring real time data visualization for academic manangemnt	DIWANSHU HIRANWAL

Strategic recommendations and intervention plans.	Develop data driven strategies for academic improvement and support mechanism	JOYAB ALAM
Resources required	Internal: Student marks dataset from GitHub, CRM data access, stakeholder interviews · External: ABS education statistics, Power BI, Python, public education datasets	
Preliminary System Requirements	Functional: Dashboard must show averages (with/without FNE/FNS), performance trends, course comparisons · Non-functional: Responsive design, intuitive navigation, fast loading · User Stories: "As AIH management, I want to see student performance trends to identify at-risk cohorts"	
Design and Modelling Tools (If applicable)	**Power BI, Python (pandas, Plotly), Excel, GitHub**	

Timeline

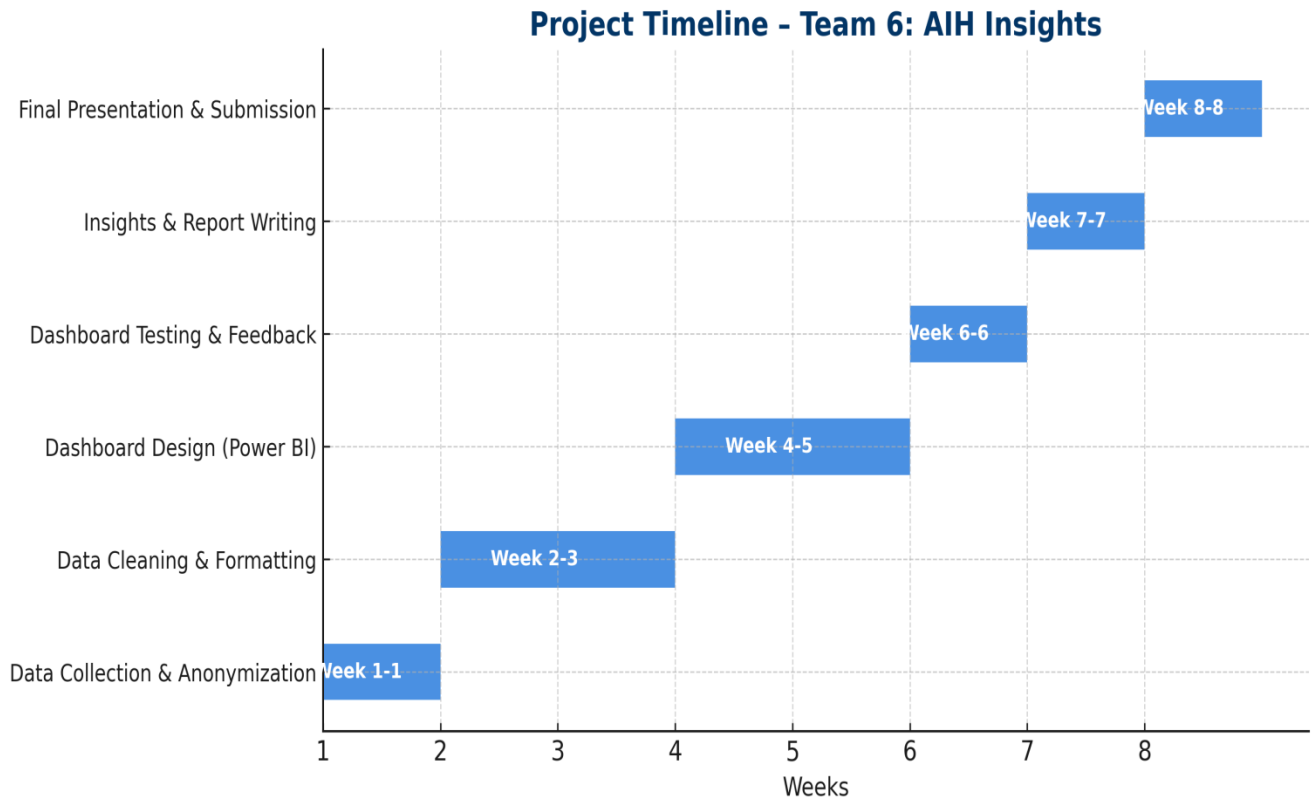

Figure 1. 8-Week Timeline – Capstone Project Delivery Schedule

Meeting Dates/Times/Venues

Client Meetings:
Week 1: Introduction and project briefing
Week 4: Webpage application progress
Week 6: Explaining our work and meeting client expectations
Week 8: Submission of report and presentation.

Communication Channels

For better communication the team and our client collectively decided to use Zoom meeting as our primary source of communication. Moreover, we can communicate through outlook mail as well Practera chat box.

Project Management Tools Used

For task tracking and controlling we are using Github. For designing the webpage application and data visualization we are using PowerBi. Excel is used for anonymizing datasets.